Deployment- PACKING LIST

Operator Name: Krish Hale

Date: 06/20/2025

Essential Tools:

- [x] Cordless power drill and extra batteries (charged)
- [x] Flathead socket driver ¼"
- [x] Flathead screwdriver ¼"
- [x] Phillips head screwdriver
- [x] Precision screwdriver (for small screw terminals)
- [x] Large diagonal cutters and large needle nose pliers (for trimming excess hose clamp material)
- [x] Utility cutter or small diagonal cutters (for cutting zip ties)
- [x] Cart
- [x] PPE
- [x] Work Gloves

Installation Components:

- [ ] Signs (number to be installed) — didnt locate contact mailroom
- [ ] Hose clamps – 2 per box plus 2 per sign
- [x] UV-resistant tape
- [x] Scissors to cut tape
- [x] UV-resistant zip ties (assorted sizes)
- [x] Dessicant
- [ ] Tape

Deployment Units:

- [x] Primary deployment sensor boxes (number to be installed) 11 done due to car delays
- [x] Spare/extra deployment boxes – Qty 3 did not carry spare

Clamps for clipboard

Marking & Orientation and Checklists:

- ☑ Permanent marker (Sharpie) + extras
- ☑ Magnetic compass (for ensuring correct southern orientation of boxes)
- ☑ Clipboard with 1 set of packing list and # installation checklists based on number to be installed on the respective day

Deployment- PROCEDURE LIST

Operator Name: ____________________________

Date: ________________

Sensor Box Mounting Tasks

Device Placement

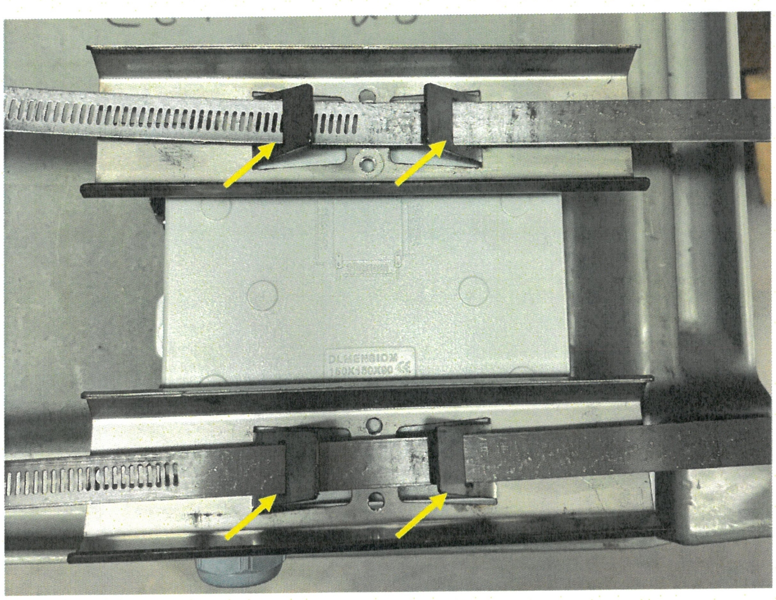
- ☐ Transport the pre-assembled device to the installation site.

- ☐ Place the box on its clamp side (the side with the box closing latches).
  Caution: Do not place the box on the side with the solar shield to avoid potential damage to external components.

Mounting Procedure

- ☐ Position the box on the pole at a height of 10 feet to the bottom of the box and ensure that it faces true south (approximately 14 degrees offset from magnetic south at 194 degrees) using the compass.

- ☐ Secure the box using two hose clamps by passing them through the cross members as shown in the picture:



- ☐ Lightly tighten the first hose clamp to hold the box in place.

- [ ] Partially tighten the second clamp so the box is close to secure but can still be rotated to correct southern facing position.

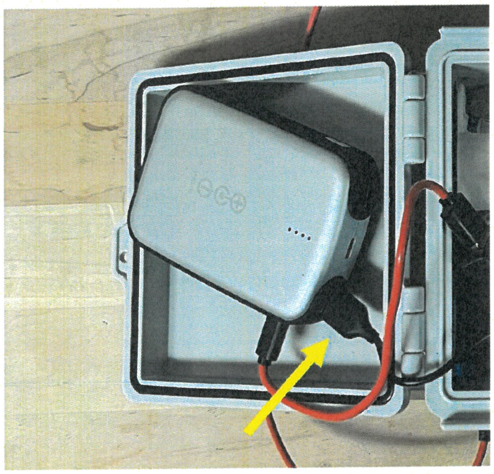
- [ ] Once both clamps and the solar panel are positioned correctly, fully tighten both to secure the box firmly.

- [ ] Clip the excess hose clamps using a cutter.

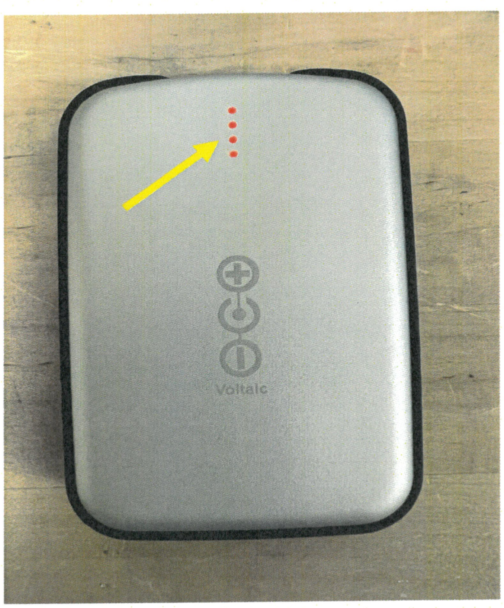
Sensor Box Startup Verification

Power Up

- ☐ Connect the USB cable to the power bank, use the port as shown in the picture.

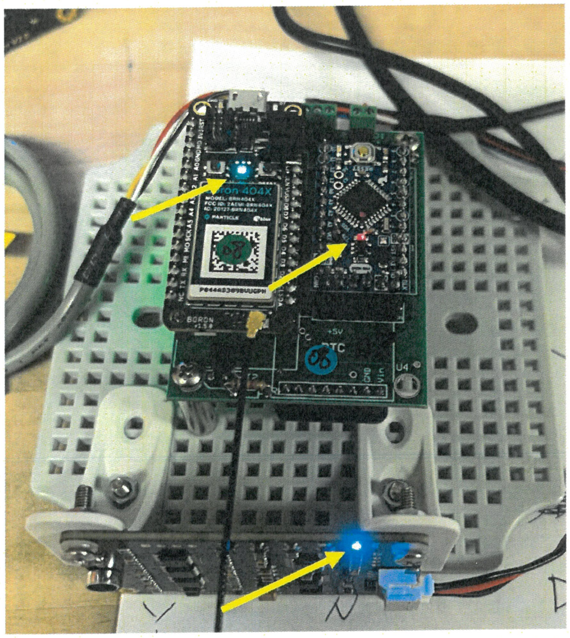


- ☐ Press the button on the power bank to verify it has full charge. 4/4 points will light up and the red indicator for the arduino will turn on as shown in the picture.
  Note: in some cases due to the winter the points take some time to show up or don't show up at all. In this case proceed with the visual indicator check to confirm that the Arduino is getting power and then look for the sound sensor and LTE module lights.



Visual Indicator Check

- Confirm that all indicator lights are functioning:



- Indicator Light Confirmation:

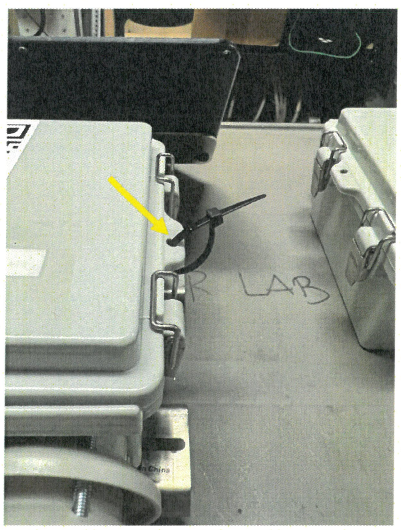
  - ☐ Arduino Green Light: ☐ Yes ☐ No
  - ☐ Sound Sensor Blue Light: ☐ Yes ☐ No
  - ☐ LTE Module Light Blue Light: ☐ Yes ☐ No
  - ☐ Data Upload Confirmation (Particle.io): ☐ Successful ☐ Not Successful (Retrieve box and use the spare boxes that are brought)

Data Transmission Verification

- Log into the particle io website to look for the startup message the module sends when it is powered up to confirm that the cellular module is uploading data. Verify that the Box number matches the Box number in the message as shown by the picture..

Box Closing

- ☐ Check that no electronic wires are caught or pinched before attempting to close the door. **Note:** Improper closure can seriously affect internal components and cause failures.
- ☐ Close the box door and secure it by threading a zip tie through the hole located on the outer edge of the box as shown in the picture.



Important: While closing, ensure the power cable and the yellow solar panel charging cable is not interfering with the box door.

Installation Checklist Recording

- GPS Location: ____________
- Box Number: ____________
- Time of turning on the box after Installation: ________________
- Battery Voltage # of Points: ________________
- Photograph taken☐ Yes ☐ No

Deployment- PROCEDURE LIST

Operator Name: Krish Hute
Date: 6/26

Sensor Box Mounting Tasks

Device Placement

- [x] Transport the pre-assembled device to the installation site.
- [x] Place the box on its clamp side (the side with the box closing latches).
  Caution: Do not place the box on the side with the solar shield to avoid potential damage to external components.

Mounting Procedure

- [x] Position the box on the pole at a height of 10 feet to the bottom of the box and ensure that it faces true south (approximately 14 degrees offset from magnetic south at 194 degrees) using the compass.
- [x] Secure the box using two hose clamps by passing them through the cross members.
- [x] Lightly tighten the first hose clamp to hold the box in place.
- [x] Partially tighten the second clamp so the box is close to secure but can still be rotated to correct southern facing position.
- [x] Once both clamps and the solar panel are positioned correctly, fully tighten both to secure the box firmly.
- [x] Clip the excess hose clamps using a cutter.

Deployment- PROCEDURE LIST

Sensor Box Startup Verification

Power Up

☑ Connect the USB cable to the power bank.

☑ Press the button on the power bank to verify it has full charge. 4/4 points will light up and the red indicator for the arduino will turn on.
Note: in some cases due to the winter the points take some time to show up or don't show up at all. In this case proceed with the visual indicator check to confirm that the Arduino is getting power and then look for the sound sensor and LTE module lights.

Visual Indicator Check

- Confirm that all indicator lights are functioning:
- Indicator Light Confirmation:

  ☑ Arduino Red Light: ☑ Yes ☐ No

  ☑ Sound Sensor Blue Light: ☑ Yes ☐ No

  ☑ LTE Module Light Blue Light: ☑ Yes ☐ No

  ☑ Data Upload Confirmation (Particle.io): ☑ Successful ☐ Not Successful
  (Retrieve box and use the spare boxes that are brought)

Data Transmission Verification

- Log into the particle io website to look for the startup message the module sends when it is powered up to confirm that the cellular module is uploading data. Verify that the Box number matches the Box number in the message.

Box Closing

☑ Check that no electronic wires are caught or pinched before attempting to close the door.
Note: Improper closure can seriously affect internal components and cause failures.

☑ Close the box door and secure it by threading a zip tie through the hole located on the outer edge of the box.
Important: While closing, ensure the power cable and the yellow solar panel charging cable is not interfering with the box door.

Deployment- PROCEDURE LIST

Installation Checklist Recording

- GPS Location: 42.3190030, -71.0699431
- Box Number: 7
- Time of turning on the box after Installation: 10:36
- Battery Voltage # of Points: 4/4
- Photograph taken ☑ Yes ☐ No
- Location: Kroc Center

Notes:

Deployment- PROCEDURE LIST

Operator Name: Krish Hure
Date: 6/26

Sensor Box Mounting Tasks

Device Placement

- [x] Transport the pre-assembled device to the installation site.
- [x] Place the box on its clamp side (the side with the box closing latches).
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- [x] Position the box on the pole at a height of 10 feet to the bottom of the box and ensure that it faces true south (approximately 14 degrees offset from magnetic south at 194 degrees) using the compass.
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- [x] Partially tighten the second clamp so the box is close to secure but can still be rotated to correct southern facing position.
- [x] Once both clamps and the solar panel are positioned correctly, fully tighten both to secure the box firmly.
- [x] Clip the excess hose clamps using a cutter.

Sensor Box Startup Verification

Power Up

- ☑ Connect the USB cable to the power bank.
- ☑ Press the button on the power bank to verify it has full charge. 4/4 points will light up and the red indicator for the arduino will turn on.
  Note: in some cases due to the winter the points take some time to show up or don't show up at all. In this case proceed with the visual indicator check to confirm that the Arduino is getting power and then look for the sound sensor and LTE module lights.

Visual Indicator Check

- Confirm that all indicator lights are functioning:
- Indicator Light Confirmation:
  - ☑ Arduino Red Light: ☑ Yes ☐ No
  - ☑ Sound Sensor Blue Light: ☑ Yes ☐ No
  - ☑ LTE Module Light Blue Light: ☑ Yes ☐ No
  - ☑ Data Upload Confirmation (Particle.io): ☑ Successful ☐ Not Successful
    (Retrieve box and use the spare boxes that are brought)

Data Transmission Verification

- Log into the particle io website to look for the startup message the module sends when it is powered up to confirm that the cellular module is uploading data. Verify that the Box number matches the Box number in the message.

Box Closing

- ☑ Check that no electronic wires are caught or pinched before attempting to close the door.
  Note: Improper closure can seriously affect internal components and cause failures.
- ☑ Close the box door and secure it by threading a zip tie through the hole located on the outer edge of the box.
  Important: While closing, ensure the power cable and the yellow solar panel charging cable is not interfering with the box door.

Deployment- PROCEDURE LIST

Installation Checklist Recording

- GPS Location: 42.3166749, -71.0707997
- Box Number: 8
- Time of turning on the box after Installation: 10:54
- Battery Voltage # of Points: 4/4
- Photograph taken☑ Yes ☐ No
- Location: Magnolia @ Lingard

Notes: